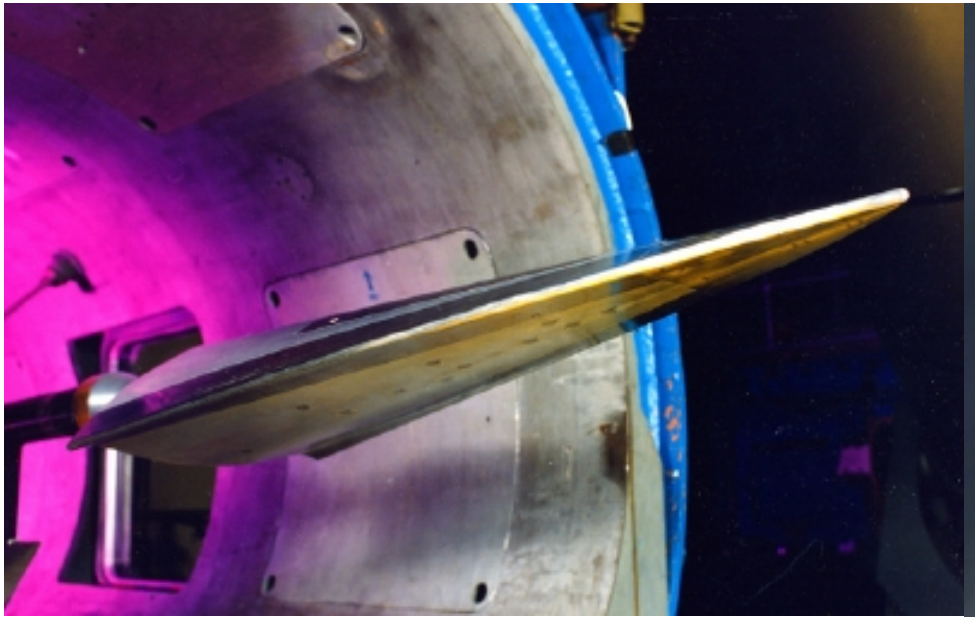
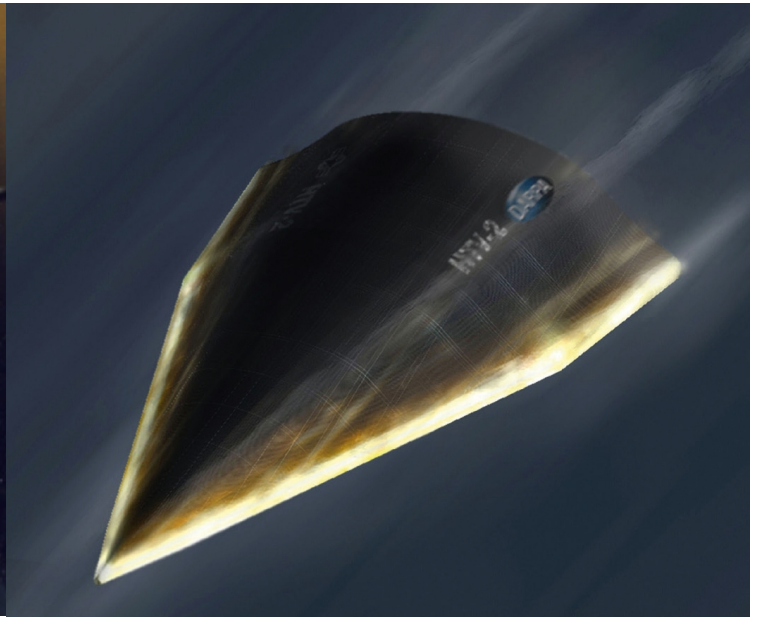
RAFAEL

Validation of the Extensions Implemented to the Simple Analytical Model for Waverider Design





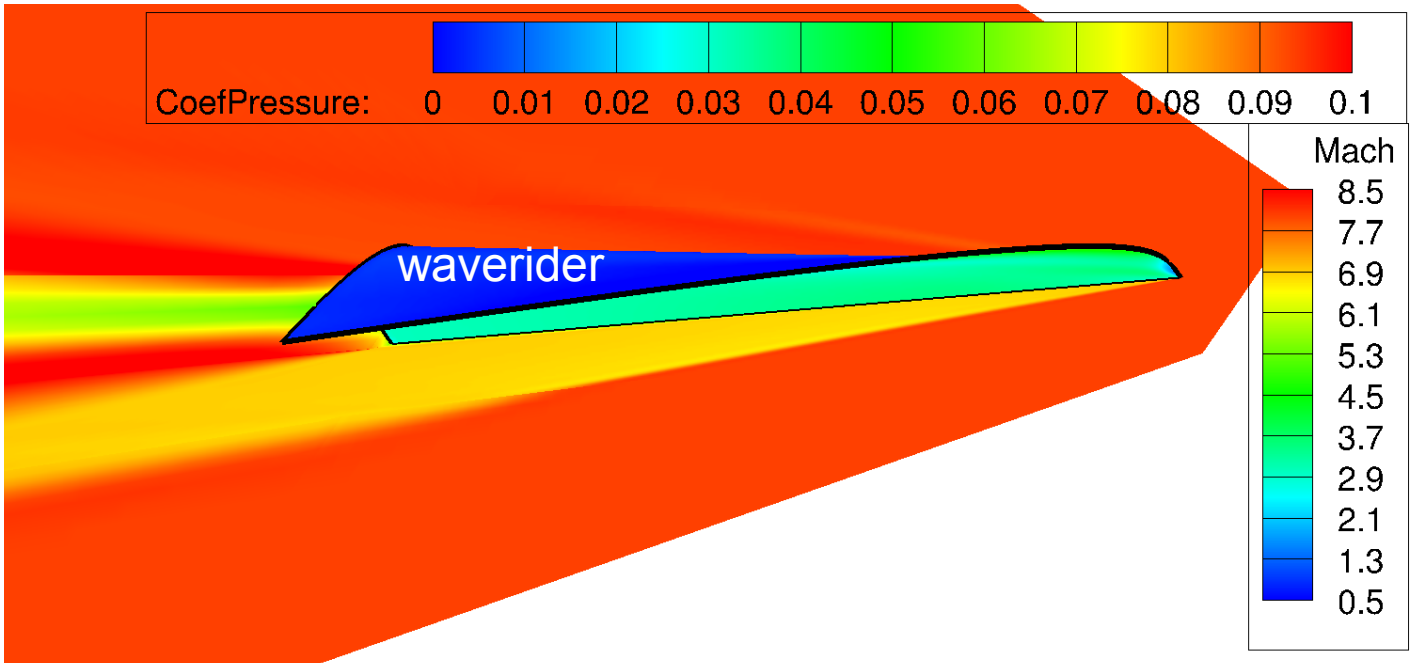
Dan Igra
CFD group
Aerodynamics Dept.

Outline

- Introduction
- Simple Analytical Model for Waverider Design
- Extensions
- Validation Cases
- Conclusions

Introduction – What is a Waverider

A waverider is a geometry that has a shock wave attached to its leading edge, this creates high pressure at its lower surface thus improving its supersonic lift-to-drag ratio (L/D). This shape is designed to “seat” on the shock wave.



Flow field around a waverider at M=8. Mach number shown at symmetry plane. Pressure coefficient shown on the waverider body.

Caret Wing Waverider

First introduced by Nonweiler in 1959

Based on 2D oblique shock wave relations

Pros: Easy to design

Cons: Low volume

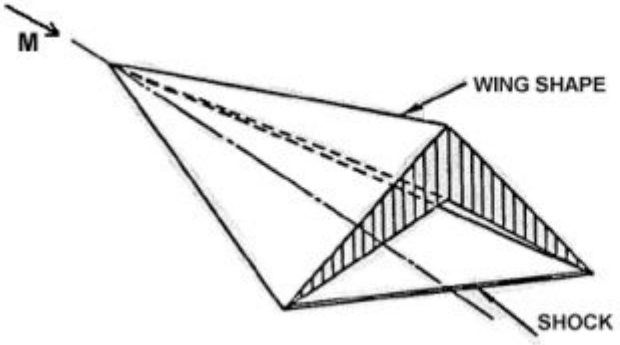


Figure taken from http://www.aerospaceweb.org/design/waverider/waverider.shtml

Conical Flow Waveriders

Based on a cone flow at high Mach numbers. Can be solved numerically by Taylor-Macoll equation or by hypersonic small disturbances theory (HSDT)

Pros: High volume

Cons: More difficult to design

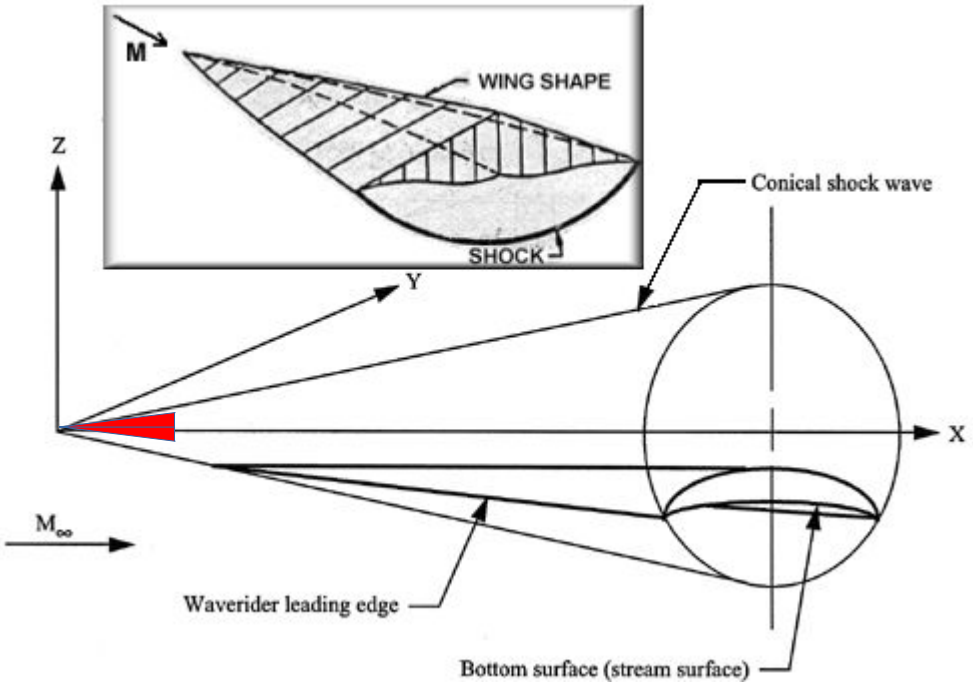
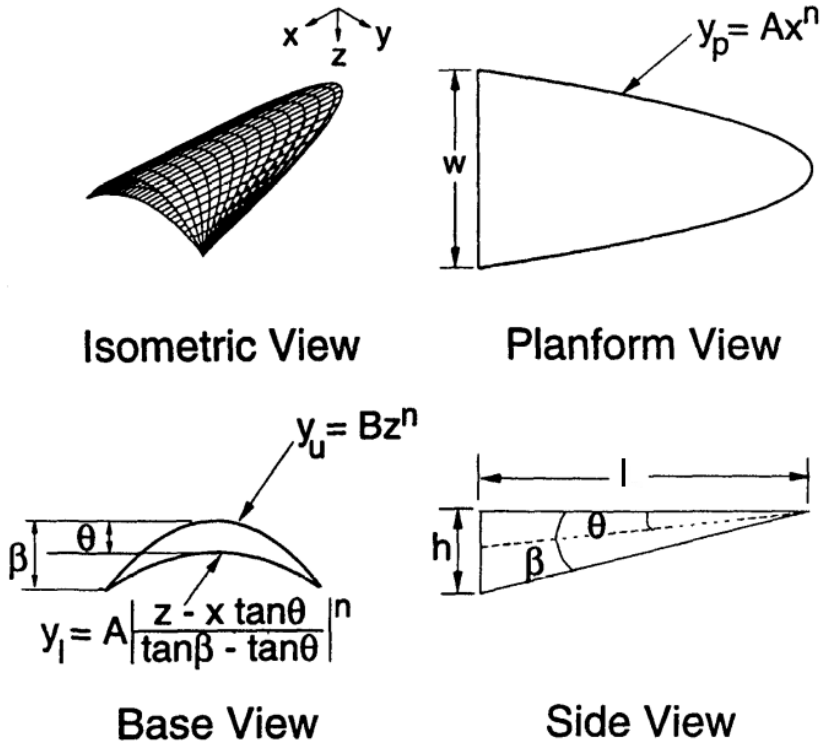


Figure taken from http://www.aerospaceweb.org/design/waverider/waverider.shtml

Simple Analytical Model for Waverider Design

This method was created by Starkey and Lewis in 1999.
Its geometry is described by equations.
It is a method for designing waveriders.
Based on these equations it is possible to compute the surfaces.



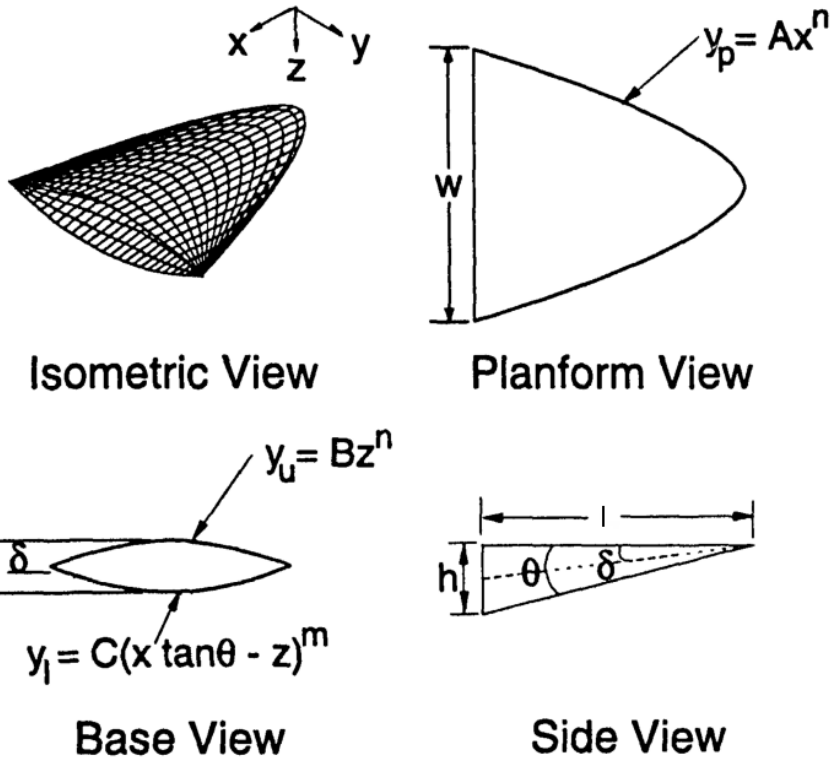
constant wedge angle waverider (CW)

Figure taken from Starkey and Lewis 1999

Simple Analytical Model for Waverider Design

Based on the Mach number, angle of attack and wedge angle it is possible to compute pressures on the surfaces based on oblique shock relations.

In the variable wedge (VW) case the waverider is treated as a series of wedges.



variable wedge angle waverider (VW)

Figure taken from Starkey and Lewis 1999

Extensions to the Simple Analytical Model for Waverider Design

The assumptions for calculating the pitch moment are:

1. The shock or expansion waves begin at the leading edge of the waverider and continue at the same initial deflection angle. This results in a uniform pressure along the waverider surface.

2. The flow is considered to be an ideal gas with constant γ.

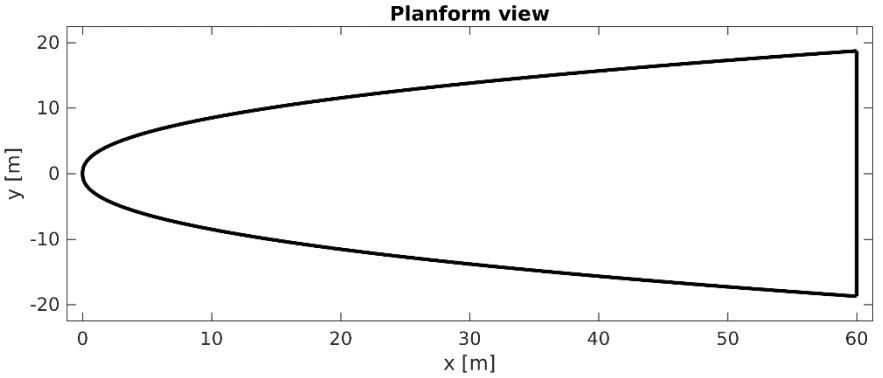
Extensions to the Simple Analytical Model for Waverider Design – Base Pressure Correlation

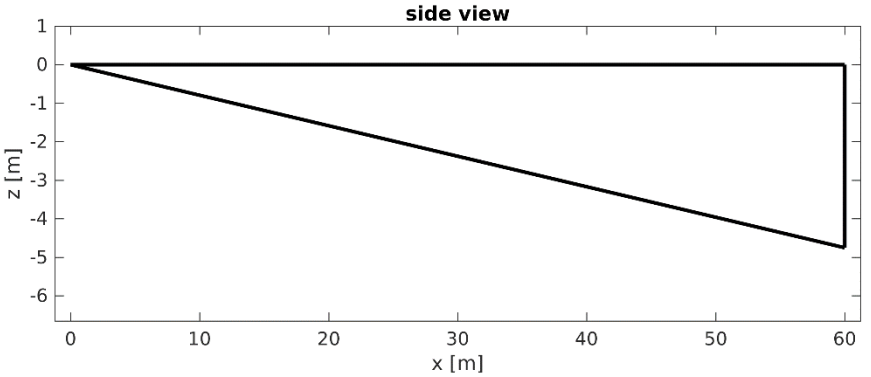
The base pressure correlation was taken from Guidi et. al 2003. It is based on the free stream Mach number for supersonic flows.

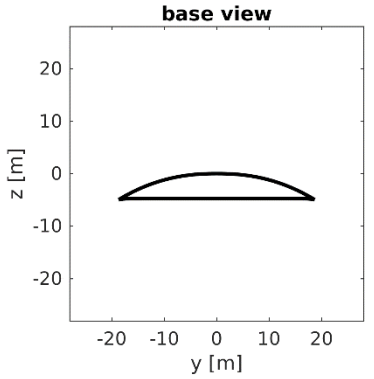
$$c_{p_{base}} = \frac{2}{\gamma M_\infty^2}\left[\left(\frac{2}{\gamma+1}\right)^{1.4}\left(\frac{1}{M_\infty}\right)^{2.8}\left(\frac{2\gamma M_\infty^2-(\gamma-1)}{\gamma+1}\right)-1\right]$$

Validation Case 1

This waverider was designed for flight at Mach number 8 at an altitude of 22 km. Geometrical information and results were taken from Starkey and Lewis 1999.







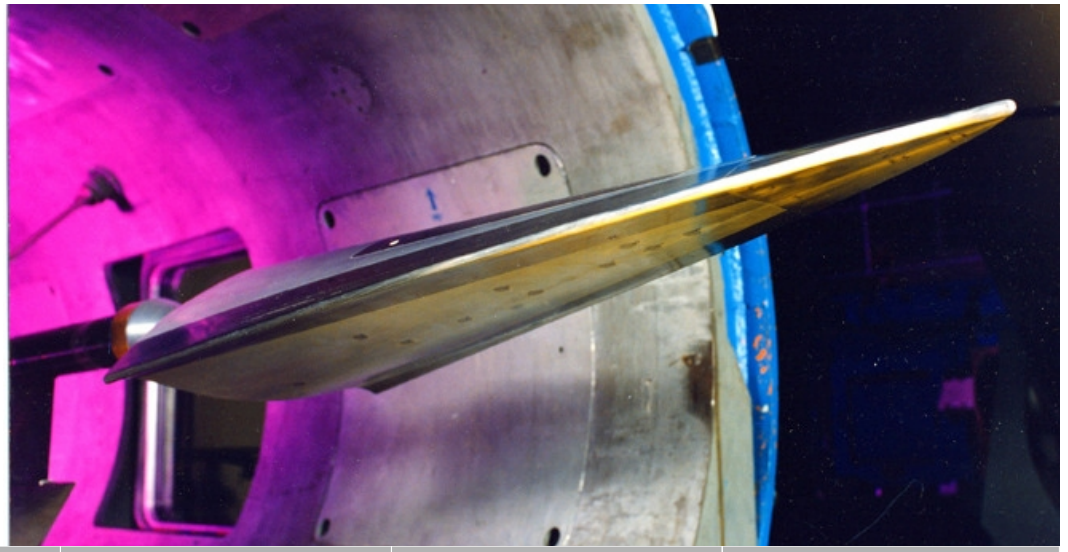
Validation Case 1

This waverider design parameters are: l=60 [m], w=37.43 [m], $\theta=4.532^0$, $\delta=4.74^0$, m=0.05 and n=0.44. Initially it was designed by MAXWARP (Maryland Axisymmetric Waverdier Program) assuming that $p_{base}=p_\infty$.

	MAXWARP	analytical $p_{base}=p_\infty$	analytical $p_{base}=f(M)$	CFD
C_D	0.0453	0.0457	0.0642	0.0614
C_L	0.3574	0.3617	0.3617	0.354
L/D	7.88	7.92	5.64	5.76

Validation Case 2

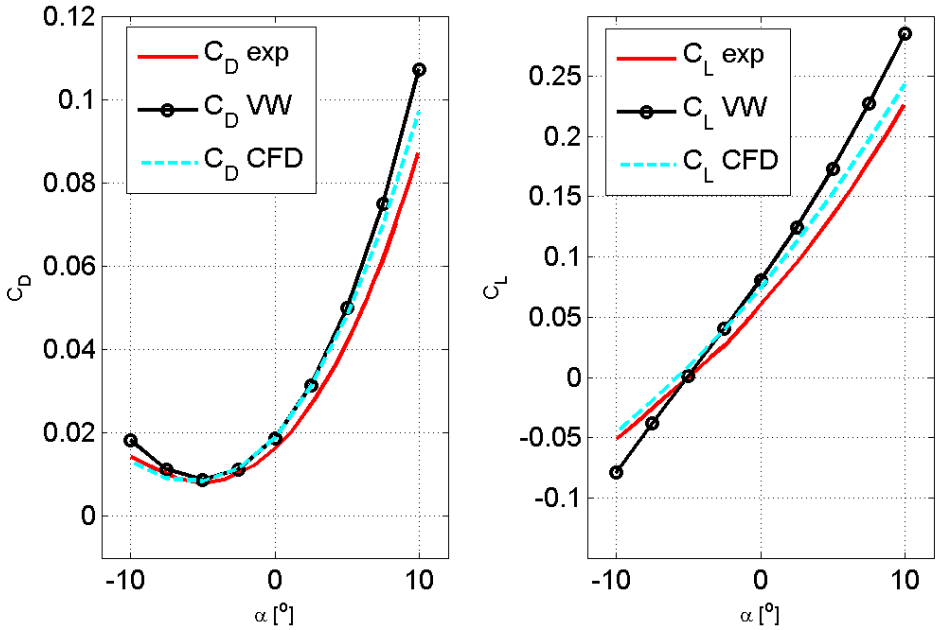
This Waverider was designed with the MAXWARP code to fly at Mach number 14. Experiments were conducted in AEDC tunnel 9, AIAA 2006-2815

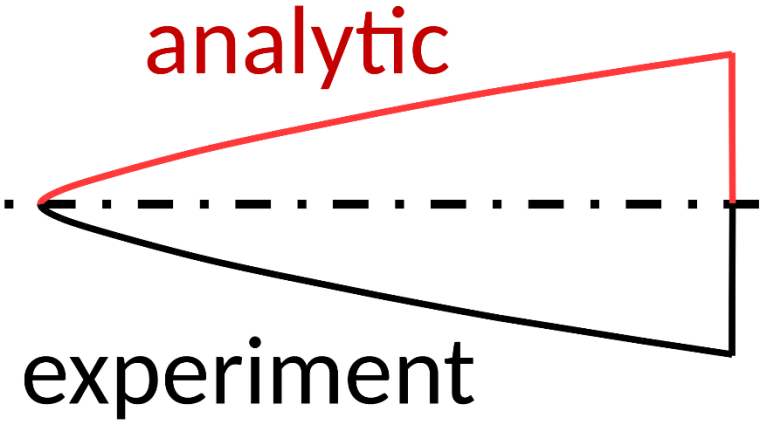


Experiment number	Mach number	Total pressure [psi]	Total temperature [K]
2389	14	9797	1707
2920	8	5040	970

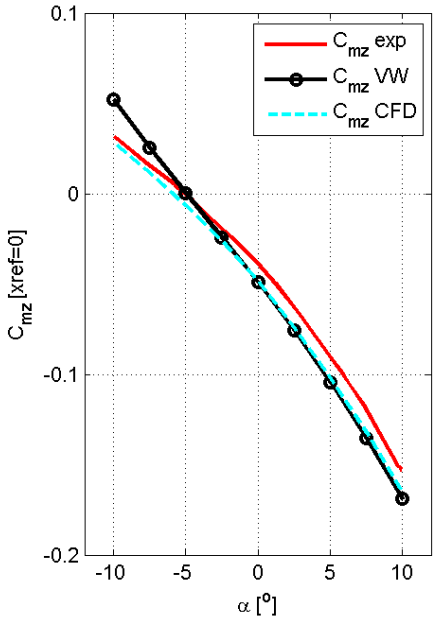
waverider design parameters are: l=39”, w=16”, $\theta=10^0$, $\delta=9^0$, m=0.6513 and n=0.7215

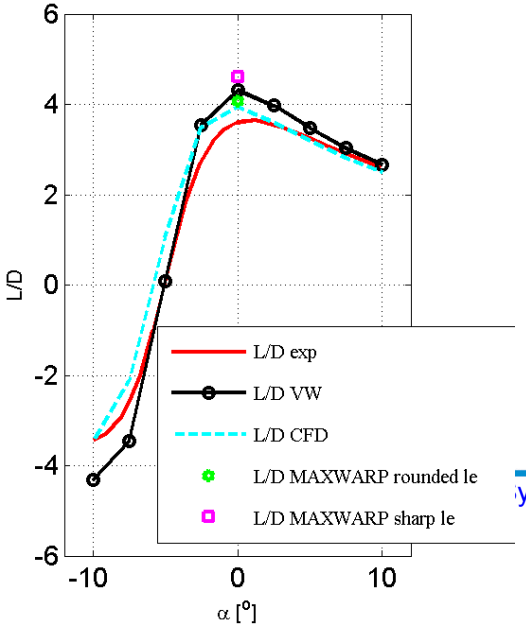
Validation Case 2

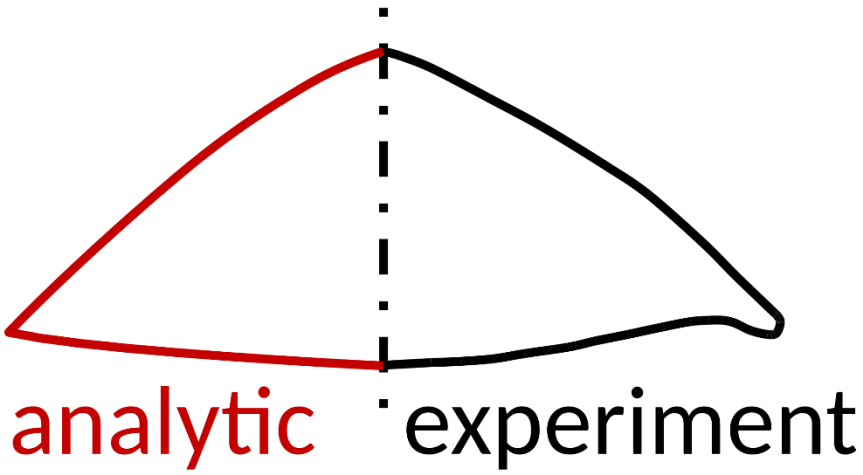




Results for Mach number 14





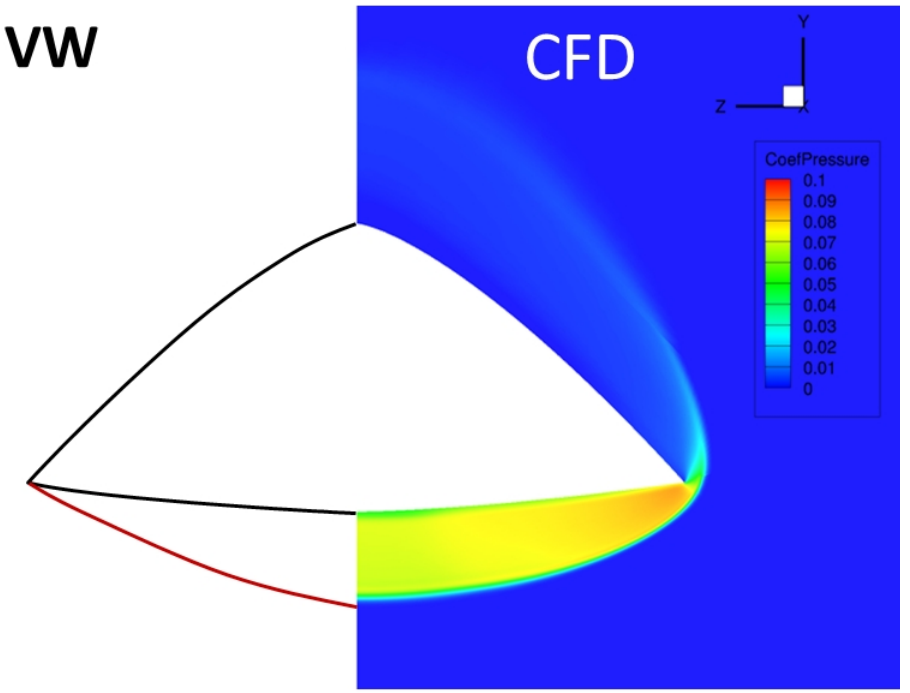


ystems LTD.

Validation Case 2

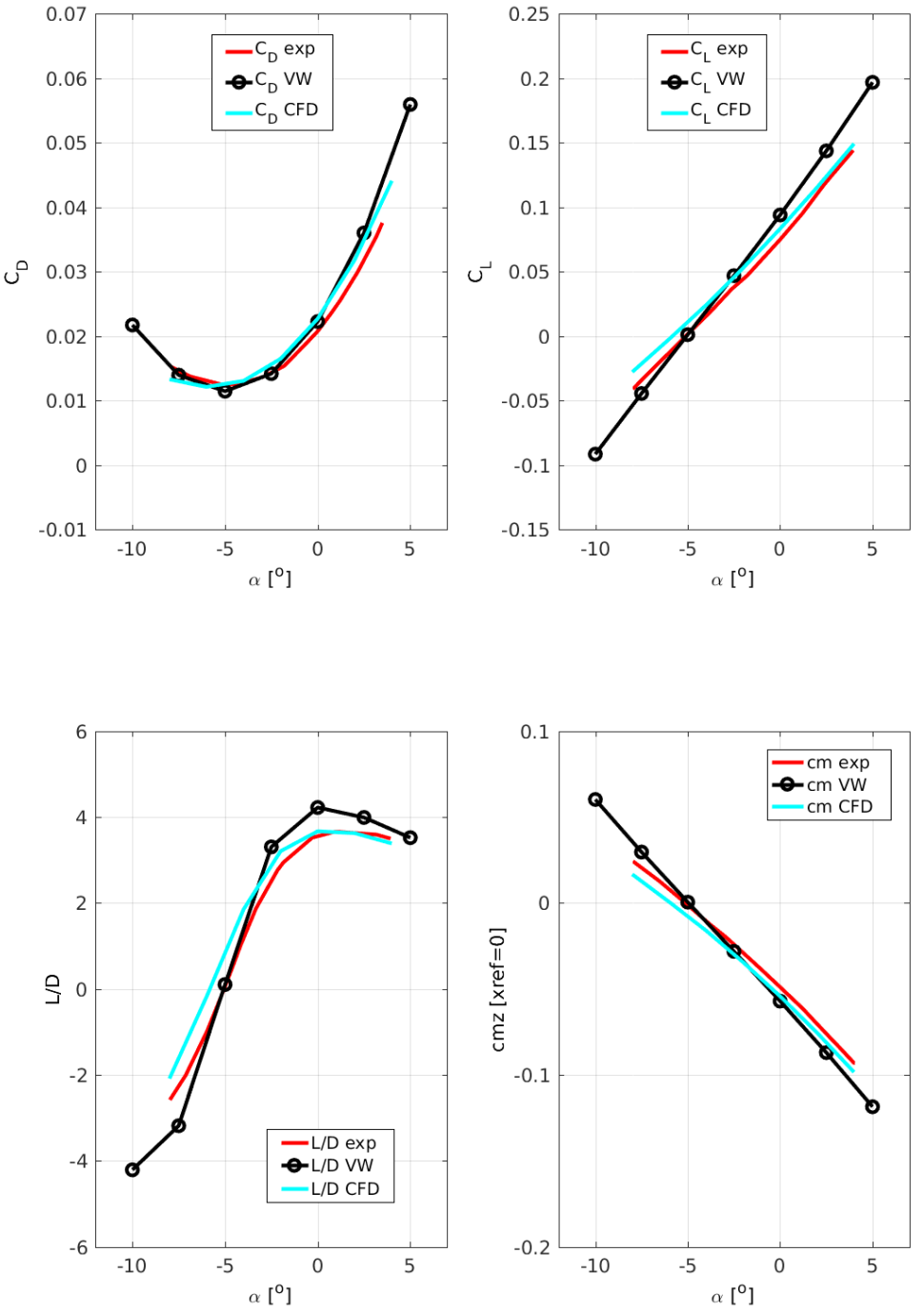
Comparison between analytically obtained shock wave location and the pressure coefficient computed via CFD, at the base of the waverider, at zero angle of attack.
Comparison of L/D computed and measured.

Data source	L/D
Experiment	3.7
MAXWARP sharp leading edges	4.61
MAXWARP round leading edges	4.07
analytical $p_{base}=f(M)$	4.31
analytical $p_{base}=0$	4.278
analytical $p_{base}=p_{\infty}$	4.59
CFD	3.94



Validation Case 2

Comparison for the case at off design Mach number, at Mach number 8.

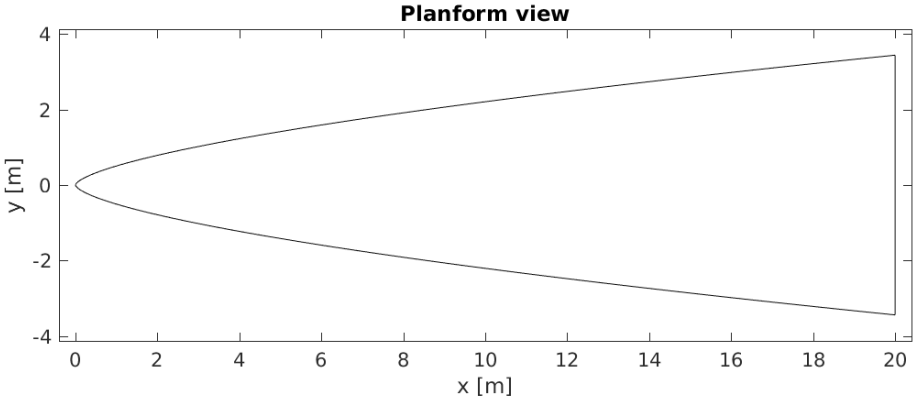


Validation Case 3

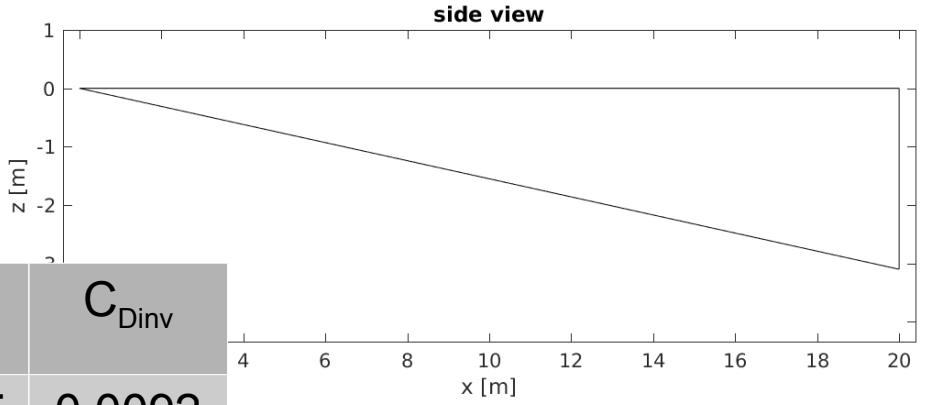
- This waverider was taken form He & Emanuel 1994.
- Flying at Mach number 8 at an altitude of 24 [km], for these conditions the Reynolds number based on the length of the waverider is $1.55*10^{8}$.
- In He & Emanuel this waverider was computed with the HSDT (hypersonic small disturbance theory) with a laminar boundary layer approximation.
- The design parameters were converted to those of the simple analytical model for waverider design, they are: l= 20 [m], $\theta=8.81^{0}$, $\delta=10.8^{0}$, n=0.641, m=0.12 and w=6.88 [m].

Validation Case 3

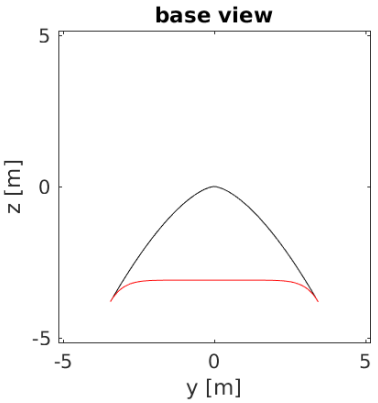
The geometry after the conversion to the simple analytical model for waverider design.



Comparison of L/D for Mach number 8

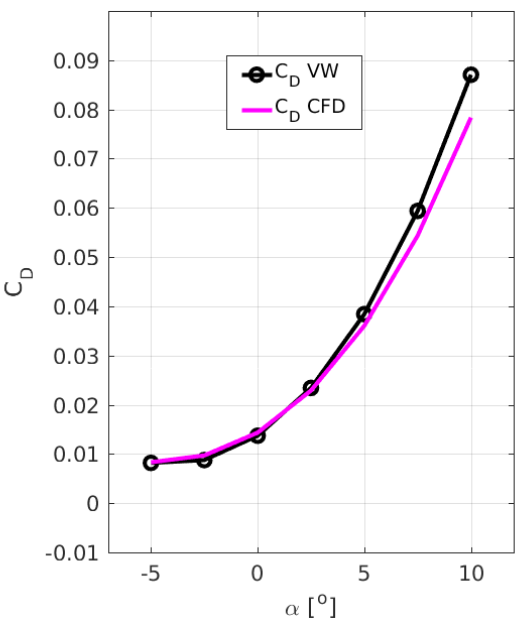


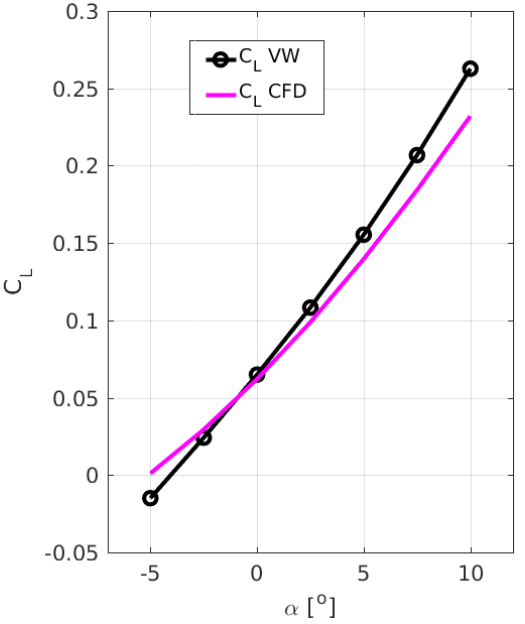
	L/D	L/D_{inv}	C_L	C_D	C_{Dinv}
Results from He & Emanuel 1994	7	7.26	0.0668	0.0095	0.0092
Current results $p_{base}=p_\infty$	5.82	6.99	0.0649	0.0111	0.0093
Current results $p_{base}=f(M_\infty)$	4.74	5.48	0.0649	0.0137	0.0117

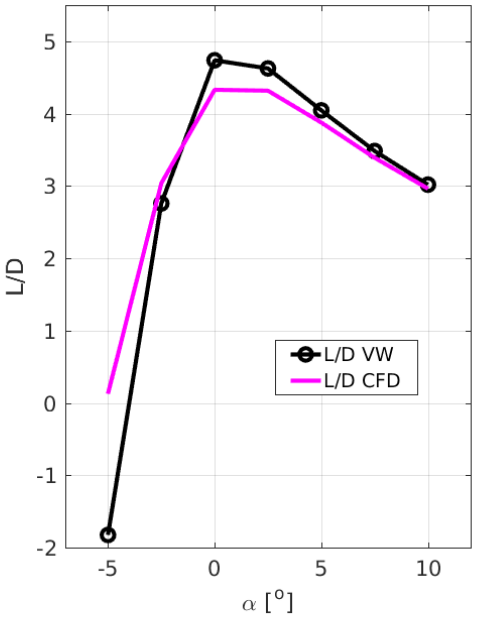


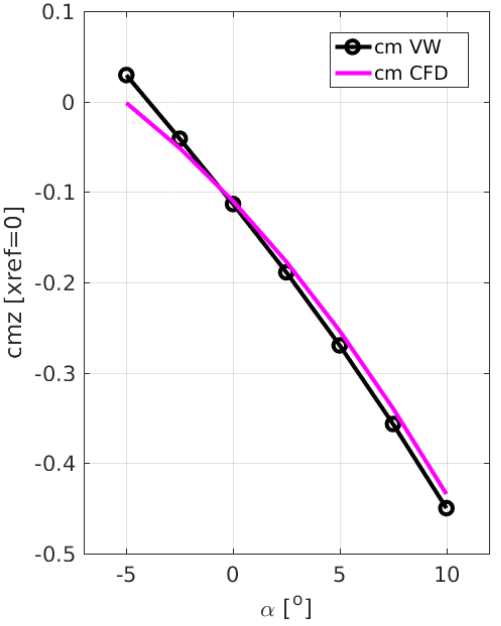
Validation Case 3

Comparison between the aerodynamic coefficients obtained by the analytical method and CFD computed at Mach number 8.









Conclusions

- The suggested improvements to the simple analytical model for waverider design demonstrated the ability of the variable wedge method to predict accurately the aerodynamic properties for all three different cases presented.
- This method has been extended in two ways; one is the calculation of the pitch moment and the other is estimation of the base pressure.
- Employing these additions resulted in a very good agreement with both experimental and CFD results.
- The importance of the base pressure estimation has been demonstrated. In spite of the high Mach numbers it has a large influence on the drag and on L/D ratio.